GRAPH-STRUCTURED MEMORY FOR COGNITIVE AGENTS

FGWM @ KI 2026

From Knowledge Graphs to Non-Euclidean Geometry

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Overview

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Why Graph Memory?

The Limits of Flat Memory

- LLM agents need memory that spans sessions, tracks change, and supports multi-hop reasoning
- Dominant approaches: vector stores, key-value extraction, or brute-force context compression
- These work for factual recall — but fail systematically on:
    - Temporal reasoning — what changed and when?
    - Multi-hop inference — connecting facts across steps
    - Knowledge evolution — updating beliefs over time

Benchmark Evidence

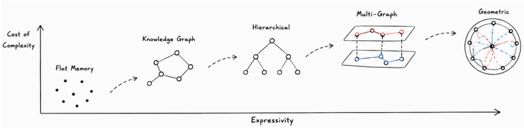
Cognitive accuracy below 25% across Mem0, Letta, and Zep (LoCoMo-Plus, 2026).
Multi-hop conflict resolution below 7% for all methods (MemoryAgentBench).

Graph Memory: A Different Strategy

Instead of compressing everything:

- Store knowledge in a graph
- Retrieve selectively via topology
- Entities and relations are first-class
- Temporal intervals on edges

Multi-strategy graph systems achieve 91.4% retrieval vs. 49.0% for extraction-based (LongMemEval).



Progression from flat memory to geometric representations.

02

Graph Memory Taxonomy

Six Architectural Patterns

Established:

1. **Entity-relation KGs** — typed triples (GraphRAG, CraniMem, AriGraph)
2. **Hierarchical event graphs** — episodic + semantic layers (GAM, MemFly, MemGraphRAG)
3. **Bipartite / associative** — lightweight linking (LP-RAG, AssoMem)
4. **Multi-graph composites** — orthogonal views (MAGMA, TM2RAG)

Emerging:

5. **Hypergraph memory** — *n*-ary facts (HyperGraphRAG, HyperMem)
6. **Procedural strategy graphs** — agent trajectories (TGM, PRAXIS, SkillRL)

Key Finding

Entity-relation KGs account for 60% of surveyed systems. Non-KG structures remain rare.

The Temporal Grounding Deficit

- Despite being the central promise, 57% of surveyed systems implement no temporal mechanism
- Only 20% go beyond simple timestamps
- Best solution: Graphiti/Zep with bi-temporal model (event time vs. ingestion time, validity intervals on every edge)

What exists:

- RecallM — temporal windows
- MemoTime — time trees
- Mnemosyne — exponential decay

What's missing:

- Staleness tracking for semantic knowledge
- Continuous-time evolution
- Principled temporal fusion

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Retrieval Spectrum

From Graph Algorithms to Learned Retrieval

- Training-free (dominant): PageRank, Leiden, recency decay — 10–150 ms, works on any graph
- Multi-strategy: semantic + BM25 + graph traversal + temporal, fused via cross-encoder (91.4% on LongMemEval)
- RL-based: learned edge weights, memory operations, graph construction
- GNN message passing: only a handful of systems (cold start and evolution barriers)

The Model-Size Equalizer

Graph memory disproportionately benefits smaller models. GAM: +39% F1 at 7B, only +7% at GPT-4o. Trained 4B with graph memory outperforms baseline 8B.

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The Geometric Frontier

Why Geometry Matters

The problem:

- All surveyed systems embed graphs in flat Euclidean space
- But memory graphs are inherently hierarchical and tree-like
- Euclidean volume grows polynomially —
  tree branching grows exponentially
- Result: semantic collapse as memory grows

The intuition:

- Hyperbolic space has exponential volume growth — naturally separates hierarchy levels
- Abstract topics near the center, specific facts near the boundary
- Curvature diagnostics flag retrieval bottlenecks without training
- Already proven in knowledge graph reasoning

Recent evidence: LLM token embeddings already exhibit negative Ricci curvature — suggesting LLM representations are naturally hyperbolic (HELM, 2025).

Three Geometric Techniques

1. Discrete Ricci curvature — per-edge diagnostic
   Flags bottleneck edges causing over-squashing. Training-free, $O(|E|)$.
2. Hyperbolic embeddings — hierarchy-preserving retrieval
   Drop-in replacement for Euclidean representations. HyperbolicRAG, HypRAG show gains.
3. Lie group manifolds — temporal knowledge fusion
   Entities, relations, timestamps in different spaces — Lie algebra enables principled fusion.

Plus: mixed-curvature product spaces for heterogeneous graphs (GraphMoRE).

Concrete Example: Enhancing Graphiti/Zep

Starting point: strongest bi-temporal system in our survey

Entity-relation KG with validity intervals on every edge.

1. Curvature diagnostic — Forman-Ricci curvature flags retrieval bottlenecks automatically
   No training needed. Explains the cognitive accuracy ceiling.
2. Hyperbolic embeddings — Poincaré ball replaces Euclidean entity vectors
   Abstract topics near center, specific facts near boundary.
3. Temporal fusion — $SO(2)$ rotations fuse validity intervals with embeddings
   Answer *what*, *when*, and *how beliefs evolved*.

None require retraining the LLM — they operate on the memory layer.

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Takeaway

Key Takeaway

The Central Argument

The structural limitations of graph-based agent memory are consequences of a geometric mismatch. Non-Euclidean techniques directly address them.

- Graph memory works — top benchmark scores, disproportionate gains for smaller models
- Temporal grounding deficit (57% of systems) is the most urgent gap
- Geometric techniques proven in adjacent fields — several are training-free
- Open challenge: integrating into persistent, cross-session architectures

Banf, Kuhn, Garcarz, Imasheva, Filipiak, Stiller, Mader, Kalek, Schattauer, Steuer, Luczyna, Gärtner, Rasenecker, Becker, Fetz — Perelyn GmbH

Discussion

Questions & Discussion

- How does your system handle memory that evolves over time?
- Which structural limitation — multi-hop ceilings, temporal gaps, or semantic collapse — resonates most with your experience?
- Should geometric representations be built into the memory layer or remain a post-processing step?

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